

Universal Compression of Memoryless Sources over Large Alphabets

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Abstract—The problem of universal coding of memoryless sources has been studied extensively for decades and it is understood that we can achieve a compression as good as in the case of classical source coding for finite source alphabets when the number of symbols is large. In this work, we will examine some works about universal compression ([1], [2], [3]) with a special interest in the "large alphabet" case. Although it is impossible to limit the redundancy when the alphabet is large, we will see that some part of the redundancy can still be limited by exploring these papers. First, referencing [1], we will take a broader look at the universal compression problem. Next, we will suggest a method of universal compression over any arbitrary unknown alphabet referencing [2]. Later, referencing [3], we will compare various approaches to universal compression over large alphabets. Lastly, we will propose our ideas on how to improve this research.

Index Terms—Universal Compression, Side Information, Unknown Alphabets, Large Alphabets, Worst-case Redundancy, Pattern, Shape, Block Compression, Sequential Compression

I. INTRODUCTION

THE probability distribution of the source is not known for most data compression applications. These source coding problems where the source distribution is not known are called "Universal Compression". However, sometimes, it is known that the probability distribution of the source is from a certain class of distributions although the exact distribution is not known. It may even be determined which distribution in this class it is by some side information. In this work, we will study several types of universal compression problems such as universal compression with side information (by examining the different cases where this side information is available to the decoder or not), and universal compression of memoryless sources over any unknown alphabets, focusing on the large, possibly infinite, alphabets.

We examined these variations of the Universal Coding Problem by looking at the performance of two different classes of codes, namely block codes and variable-length codes. We defined the performance measure as the probability of error in block codes and redundancy (average or worst-case redundancy in different situations) in variable-length codes. We first looked at the universal compression with side information and saw that we could get a diminishing per-symbol probability of error for the block codes and a diminishing per-symbol redundancy for the variable-length codes in the asymptotical case. Then, we explained the insufficiency of some methods developed for a finite known alphabet when the alphabet is arbitrary and unknown. We explained why achieving a finite redundancy is impossible as the alphabet size grows. But, even so, we expressed a technique that makes some part of this redundancy, namely pattern redundancy, diminish. Lastly, we

examined another method for a memoryless source of some known large, possibly infinite, alphabets. We saw that some parts of the redundancy, namely shape redundancy, can be bounded.

In section II, we will look at the universal coding problem with the side information studied in [1]. This paper is important as it provides a very fundamental setup and approaches to the universal compression problem. It defines a class of distributions where the distribution of the source is coming from and describes the source distribution in this class by a random variable called side information. Then, it considers 2 different cases where the decoder has the side information or not. For both cases, it analyzes the performance of block codes and variable length codes by defining performance measures. It defines the performance of an (N, R) block code as the average error probability of the coding scheme where the decoder is the maximum likelihood decoder and the performance of a variable length code as the redundancy, the number of bits in excess of the source entropy. In the block codes case, by making a memoryless source assumption, it shows the existence of an (N, R) block code with a diminishing probability of error as N grows for sufficiently big values of R by upper bounding the average error probability of a random ensemble of (N, R) block codes. Additionally, by lower bounding the error probability for the case where the side information is both available to the encoder and the decoder, it proves the existence of an (N, R) block code that is at least as good as any code for the case where the side information is both available to the encoder and the decoder when N goes to infinity and R is in a certain range. Moreover, it defines minmax and maxmin redundancies as two different performance measures for the variable length codes. Then, it proves that maxmin and minmax redundancies are essentially the same and they are both equal to the per symbol capacity of the channel from the side information to the source.

In section III, we will look at the problem of universal compression of memoryless sources over unknown alphabets discussed in [2]. In contrast to [1], this paper deals with a more specific issue in universal compression. It starts by defining the worst-case redundancy and presenting some results in the literature about calculating it. Then, it explains why it is impossible to achieve finite redundancy over an unknown arbitrary alphabet, and it suggests limiting some part of the redundancy. With this goal in mind, it suggests expressing a sequence of strings by a pattern and a dictionary. It calls the redundancy coming from conveying the pattern of the sequence "pattern redundancy". It extensively studies patterns and finds an equivalency between patterns and set distributions. Using this equivalency, it finds some combinatorial results. Using

these results along with Shtarkov's sum [4] and some other results from the literature, it proves that it can be bounded by a sublinear function of the number of strings (symbols) in the sequence, n . Thus, per-symbol pattern redundancy diminishes as n grows. Finally, this paper provides a way to design both block and sequential compression schemes that achieve diminishing per-symbol pattern redundancy.

In section IV, we will examine the problem of universal compression of memoryless sources over large alphabets discussed in [3]. On top of the pattern approach discussed in [2], this paper ([3]) proposes two new approaches for dealing with the large alphabets, namely the standard compression approach and the shape approach. Using the results in [5], [6], [7], [8], [9], [10], [11], [12], it determines the rate at which per-symbol standard redundancy goes to infinity as the alphabet size increases. Moreover, it defines the concept of shapes, and it shows that the per-symbol shape redundancy is between 0.027 and 1. Furthermore, it studies the pattern approach in [2] by presenting a few extra results for some special cases. It points out the differences between these 3 approaches and explains the advantages on top of each other. Lastly, it presents some combinatorial interpretations using the equivalency between the shapes and ordered set partitions and the equivalency between the patterns and the unordered set partitions. In this report, we will summarize these 3 approaches in [3], and compare the results acquired in [3] for these three approaches.

In conclusion, we studied universal compression from a broad perspective by focusing on the case where the source is memoryless and its alphabet is large. We defined some performance measures and analyzed the performances of both block and variable length codes. We explained the impossibility of achieving finite redundancy for infinite alphabets, but, in the end, we showed that some part of this redundancy can be bounded by studying patterns and shapes. Lastly, we proposed some ideas about how to advance this research in section V.

Note that for the sake of brevity, throughout this report, we will omit some of the formal definitions related to the patterns and shapes, and we express them via some examples only. We believe that these definitions are easier to understand when explained via examples, but one may still refer to [2] and [3] for the formal definitions.

II. UNIVERSAL CODING WITH SIDE INFORMATION

This section is based on [1]. We will examine the problems discussed in [1] and summarize the results. It essentially considers two versions of universal coding problems with side information. In one of these versions, the side information is only available to the decoder and, in the other version, the side information is available to neither the encoder nor the decoder. Note that the case where the side information is both available to the encoder and the decoder is equivalent to the classical source coding problem.

A. Side Information Only at the Decoder

In this part, we will examine the universal coding with side information at the decoder for (N, R) block codes.

Given a particular random ensemble of (N, R) codes, given a source with probabilities $Q_N(\vec{x}|y)$ for some value of the side information Y , and given the optimal decoder for each code in the ensemble, we want to investigate the average probability of decoding error of the ensemble, $P_e(y)$, for the value of the side information, y . Note that, by optimal decoder, we meant the function $\text{dec}(m, y) = \vec{x}$ such that $\text{enc}(\vec{x}) = m$ and $Q_N(\vec{x}|y) \geq Q_N(\vec{x}'|y) \forall \vec{x}'$ such that $\text{enc}(\vec{x}') = m$. We will investigate $P_e(y)$ by the following 4 theorems given in [1]:

Theorem 1: For all $\rho \in [0, 1]$, and $M = \lceil 2^{NR} \rceil$

$$P_e(y) \leq M^{-\rho} \left[\sum_{\vec{x}_N} Q_N(\vec{x}|y)^{\frac{1}{1+\rho}} \right]^{1+\rho}$$

Note that proof of theorem 1 is given in Appendix-C. Now, we can represent theorem 1 as follows:

$$\log(P_e(y)) \leq -\rho NR + NE_s(\rho, N, y)$$

where $E_s(\rho, N, y) := \frac{1+\rho}{N} \log(\sum_{\vec{x}} Q_N(\vec{x}|y)^{\frac{1}{1+\rho}})$.

Theorem 2: $E_s(\rho, N, y)$ is a convex function of ρ for $\rho \geq 0$ and it attains 0 for $\rho = 0$. Moreover:

$$\frac{\partial E_s(\rho, N, y)}{\partial \rho} = \frac{1}{N} H(X_\rho^N|y)$$

where $H(X_\rho^N|y)$ is the entropy of the tilted distribution:

$$Q_{N,\rho}(\vec{x}|y) := \frac{Q_N(\vec{x}|y)^{\frac{1}{1+\rho}}}{\sum_{\vec{x}} Q_N(\vec{x}|y)^{\frac{1}{1+\rho}}}$$

Now, using the results of theorem 2, we will optimize the upper bound in theorem 1 by choosing ρ such that $NR = H(X_\rho^N|y)$. Note that we can do this only when R satisfies $H(X_\rho^N|y)|_{\rho=0} \leq NR \leq H(X_\rho^N|y)|_{\rho=1}$. So, we get:

$$\log(P_e(y)) \leq -H(X_\rho^N|X^N|y)$$

where $H(X_\rho^N|X^N|y)$ is the KL-divergence between $Q_{N,\rho}$ and Q_N , that is $D(Q_{N,\rho}||Q_N)$.

Moreover, by making a memoryless source assumption, we can update these results. Thus, ρ satisfies $NR = \sum_{n=1}^N H(Q_\rho(X_n|y_n))$ and we get:

$$\log P_e(\vec{y}) \leq \sum_{n=1}^N -H(X_\rho||X|y_n)$$

Now, let's consider a lower bound for the error probability for the best code constructed using the knowledge of the side information at both the encoder and decoder.

Theorem 3: For any set of transition probabilities $Q(x|y)$ and for all $\rho \geq 0$, there exists an $A < \infty$ such that for all (N, R) block code and every side information $\vec{y}$, if $NR < \sum_{n=1}^N H(X_\rho|y_n) - A\sqrt{N}$, then:

$$\log P_e(\vec{y}) \geq \sum_{n=1}^N -H(X_\rho||X|y_n) - A\sqrt{N}$$

Then, as we can see from these upper and lower bounds for $P_e(\vec{y})$, as $N \rightarrow \infty$, the exponent of the decay in error

probability per source symbol does not change concerning whether the side information is available to the encoder or not as long as it is available to the decoder.

Finally, assuming a distribution $P(y)$ on the side information Y and taking the average of error probability according to $P(y)$, that is $P_e = \sum_y P(y) P_e(y)$ where P_e is the average error probability over the whole random ensemble of (N, R) block codes, we will get the following theorem:

Theorem 4: (a) For a correlated memoryless source with single letter probabilities $P(y)Q(x|y)$ and with the side information $\vec{y}$ available at the decoder, the probability of error for a random ensemble of (N, R) codes, averaged over both sources $\vec{X}$ and $\vec{Y}$, satisfies:

$$R = H(X_\rho|Y) \text{ and } \log(P_e) \leq -NH(X_\rho Y_\rho || XY)$$

if $H(X|Y) \leq R \leq H(X_\rho|Y_\rho)|_{\rho=1}$. And, it also satisfies:

$$\log(P_e) \leq -NR + N \underbrace{\log\left(\sum_y P(y) \left[\sum_x Q(x|y)^{\frac{1}{1+\rho}}\right]^{1+\rho}\right)}_{\text{optimize when } \rho=1 \text{ if } R > H(X_\rho|Y_\rho)|_{\rho=1}}$$

if $R > H(X_\rho|Y_\rho)|_{\rho=1}$.

- (b) For any such source, there exists a finite constant A such that for all (N, R) encoders (including the encoders using the side information), and for all $\rho \geq 0$:

$$\log(P_e) \geq -NH(X_\rho Y_\rho || XY) - A\sqrt{N}$$

if $R \leq H(X_\rho|Y_\rho) - A/\sqrt{N}$

Now, to conclude this subsection, we will interpret the theorem 4 and derive our main result.

To achieve our main result, let's observe that there must be at least one code in the (N, R) random ensemble that attains an error probability less than P_e which is the average error probability over the whole ensemble. Then, this code (without using the side information $\vec{y}$) encodes the source sequences as well as the best code for the case where the side information is available to both the encoder and the decoder when $N \rightarrow \infty$ and $H(X|Y) \leq R \leq H(X_\rho|Y_\rho)|_{\rho=1}$.

B. Side Information neither at the encoder nor at the decoder

In this section, we will analyze a universal compression problem with side information where the side information is neither available to the encoder nor the decoder. We will have 2 approaches, using block codes, and using variable length codes:

1) *Using block codes:* Here is the illustration of the problem in figure 1:



Figure 1. Universal compression scheme using block codes

Our goal is to find an (N, R) block code (as the encoder) that minimizes the error probability with the optimal decoder. Note that the optimal decoder (which assumes distribution P

on the side information) decodes m into $\vec{x}$ which maximizes $w_P(\vec{x}) := \sum_y P(y) Q_N(\vec{x}|y)$ over all $\vec{x}$ such that $f(\vec{x}) = m$. Analogously to the part II-A, let's define $P_e(y, P)$ as the average error probability over the ensemble for a specific value of the side information y . Then, analogously to the theorem 1, we have:

$$P_e(y, P) \leq 2^{-NR\rho} \left(\sum_{\vec{x}} w_P(\vec{x})^{\frac{1}{1+\rho}} \right)^\rho \left(\sum_{\vec{x}} Q_N(\vec{x}|y) w_P(\vec{x})^{\frac{\rho}{1+\rho}} \right)$$

Averaging over Y with respect to $P(y)$:

$$P_e(P) \leq 2^{-NR\rho} \left[\sum_{\vec{x}} w_P(\vec{x})^{\frac{1}{1+\rho}} \right]^{1+\rho}, \forall \rho \in [0, 1]$$

Then, taking the supremum of both side and replacing $w_P(\vec{x}) = \sum_y P(y) Q_N(\vec{x}|y)$ into the equation:

$$\sup_P P_e(P) \leq \sup_P 2^{-NR\rho} \left[\sum_{\vec{x}} \left[\sum_y P(y) Q_N(\vec{x}|y) \right]^{\frac{1}{1+\rho}} \right]^{1+\rho}$$

Now, observe that $[\sum_y P(y) Q_N(\vec{x}|y)]^{\frac{1}{1+\rho}}$ is a concave function of P . Let's say it is maximized by a distribution P^* . Then, by KKT conditions, P^* satisfies:

$$\sum_{\vec{x}} Q_N(\vec{x}|y) w_{P^*}(\vec{x})^{\frac{-1}{1+\rho}} \leq \sum_{\vec{x}} w_{P^*}(\vec{x})^{\frac{1}{1+\rho}}$$

with equality $\forall y$ such that $P^*(y) > 0$

for the values of R satisfying $NR \geq \sup_P \sum_{\vec{x}} -w_P(\vec{x}) \log w_P(\vec{x})$. Then, replacing P with P^* in $P_e(y, P)$, we get:

$$P_e(y, P^*) \leq 2^{-NR\rho} \left[\sum_{\vec{x}} w_{P^*}(\vec{x})^{\frac{1}{1+\rho}} \right]^{1+\rho} \quad (1)$$

So, we said that the decoder assumes a $P(y)$ while finding an upper bound for $P_e(P)$. Then, we found the worst-case of these $P(y)$'s as $P^*(y)$. Now, considering our decoder as it assumes the distribution $P^*(y)$ on the side information, we acquired an upper bound on $P_e(y, P^*)$ that is the average error probability given the side information y over the random ensemble. This upper bound on $P_e(y, P^*)$ is very important because it does not depend on y . The fact that it does not depend on y originated from that the KKT conditions gave us an upper bound independent of y .

Now, having a bound for the average error probability over the ensemble in our hands, let's focus on showing the existence of an individual code that will perform well.

Theorem 5: Given any (N, R) , and a source with side information with the probability assignment $Q_N(\vec{x}|y)$, there exists an (N, R) block encoder mapping with an error probability for each y satisfying:

$$P_e(y) \leq \min_{0 \leq \rho \leq 1} \max_P 2^{\Gamma_N} \left[\sum_{\vec{x}} \sum_y P(y) Q_N(\vec{x}|y)^{\frac{1}{1+\rho}} \right]^{1+\rho}$$

where $\Gamma_N = -\rho(NR - \log[\log |\mathcal{Y}|]) + 1$

Note that the proof is given in Appendix-C. Thus, for the error probability, we acquired an upper bound independent of the side information y . Also, we still have a diminishing upper bound for $P_e(y)$ as $N \rightarrow \infty$ for sufficiently big R and finite $|\mathcal{Y}|$.

2) *Using variable-length codes:* Our goal is to find a variable-length code that will provide error-free compression by using as small bits as possible. We know that even for classical source coding, the encoder needs at least $H(X^N|y)$ (conditional entropy given the side information y) bits in average to convey the source symbols to the decoder for error-free decoding. Now, let's define the redundancy of a code with length function l as the number of bits that it uses in average on top of the entropy:

$$r_N(l, y) = \frac{1}{N} \left[\sum_{\vec{x}} Q_N(\vec{x}|y) l(\vec{x}) - H(X^N|y) \right]$$

Then, we define the maxmin redundancy, $\mathcal{R}_N^-$:

$$\mathcal{R}_N^- = \sup_{P(y)} \min_{l \in \mathcal{L}} \sum_y P(y) r_N(l, y)$$

where $\mathcal{L}$ is the class of length functions satisfying Kraft's inequality. Here, we are choosing the best code for each $P(y)$ and get the average redundancies. Then, nature chooses the $P(y)$ that gives the worst of these average redundancies.

Moreover, we define the minmax redundancy, $\mathcal{R}_N^+$:

$$\mathcal{R}_N^+ = \min_{l \in \mathcal{L}} \sup_{y \in \mathcal{Y}} r_N(l, y)$$

So, we are choosing the best code without knowing the side information y such that the code will have minimum redundancy for the worst-case value of y as if nature chooses the worst value of y after we decide our code.

It is shown (by Davisson, 1973) that $\mathcal{R}_N^+ \geq \mathcal{R}_N^-$. Moreover:

$$C_N \leq N\mathcal{R}_N^- \leq C_N + 1$$

where $C_N = C_N(Y \rightarrow X^N)$ is defined as the capacity of the channel from the side information to the source.

Our goal is to show $C_N \leq N\mathcal{R}_N^+ \leq C_N + 1$. Consequently, we will show that the minmax redundancy $\mathcal{R}_N^+$ diminishes as $N \rightarrow \infty$ if $\lim_{N \rightarrow \infty} \frac{C_N}{N} = 0$.

Our first step will be getting rid of the integer constraints. Let $\hat{\mathcal{L}}$ be the class of all functions $l(\vec{x})$ that maps $\vec{x}$ into non-negative real numbers. Then, define:

$$\hat{\mathcal{R}}_N^- = \sup_{P(y)} \min_{l \in \hat{\mathcal{L}}} \sum_y P(y) r_N(l, y)$$

$$\hat{\mathcal{R}}_N^+ = \min_{l \in \hat{\mathcal{L}}} \sup_{y \in \mathcal{Y}} r_N(l, y)$$

The relations between $\hat{\mathcal{R}}_N^{+,-}$ and $\mathcal{R}_N^{+,-}$ are given in [1] as:

$$\hat{\mathcal{R}}_N^{+,-} \leq \mathcal{R}_N^{+,-} \leq \hat{\mathcal{R}}_N^{+,-} + \frac{1}{N} \quad (2)$$

Now, let's present the main theorem of this part:

Theorem 6:

$$\frac{C_N}{N} = \hat{\mathcal{R}}_N^+ = \hat{\mathcal{R}}_N^-$$

$$\frac{C_N}{N} \leq \mathcal{R}_N^- \leq \mathcal{R}_N^+ \leq \frac{C_N + 1}{N}$$

Note that it suffices to show the first equality and the remaining inequality will follow by the inequality given in 2 and $\mathcal{R}_N^+ \geq \mathcal{R}_N^-$. Proof is given in [1]

As a result of theorem 6, we showed that both minmax and maxmin redundancies without the integer constraints become equal to the capacity per symbol for the channel from the side information to the source with transition probabilities $Q_N(\vec{x}|y)$.

For instance, if we consider memoryless sources of a finite alphabet, we can see that the minmax and maxmin redundancies diminish as $N \rightarrow \infty$ by theorem 6 since the capacity can be at most $\log |\mathcal{Y}|$ which is finite.

III. UNIVERSAL COMPRESSION OF MEMORYLESS SOURCES OVER UNKNOWN ALPHABETS

This section is based on [2]. The universal coding problem that we discussed in the previous section provides us with some relation between the redundancy and the capacity of the channel from the side information to the source. Unfortunately, this does not help us to examine more general universal coding problems, such as a universal coding problem that does not contain side information. In this section, we will explore other ways to develop a universal compression scheme that could work on more general cases, even when we have an unknown arbitrary alphabet. Although it is known that a universal compression scheme that achieves a finite redundancy per symbol is impossible to design when the source alphabet is arbitrarily large, [2] shows that it is possible to bound some part of the redundancy, namely pattern redundancy. It achieves bounding this pattern redundancy, with both block and sequential encoders, by a sublinear function of the number of symbols, and concludes that per-symbol pattern redundancy diminishes as the number of symbols increases.

Now, recall that the worst-case redundancy of a distribution q with respect to a source with distribution p over an alphabet $\mathcal{X}$ is given by:

$$\hat{R}(p, q) = \sup_{x \in \mathcal{X}} \log \frac{p(x)}{q(x)}$$

Then, the worst case redundancy of q with respect to a class of distributions $\mathcal{P}$ is given by:

$$\hat{R}(\mathcal{P}, q) = \sup_{p \in \mathcal{P}} \hat{R}(p, q) = \sup_{p \in \mathcal{P}} \sup_{x \in \mathcal{X}} \log \frac{p(x)}{q(x)}$$

Finally, the worst case redundancy of $\mathcal{P}$ is given by:

$$\hat{R}(\mathcal{P}) = \inf_q \hat{R}(\mathcal{P}, q) = \inf_q \sup_{p \in \mathcal{P}} \sup_{x \in \mathcal{X}} \log \frac{p(x)}{q(x)} \quad (3)$$

Then, it is stated in [2] that, the succession of papers [5], [6], [7], [8], [9], [10], [11], [12] proves that the worst case redundancy of an i.i.d. source of length n coming from an alphabet of size m can be expressed as:

$$\hat{R}(I_m^n) = \frac{m-1}{2} \log \frac{n}{2\pi} + \log \frac{\Gamma(\frac{1}{2})^m}{\Gamma(\frac{m}{2})} + o_m(1) \quad (4)$$

where Γ is the gamma function and $o_m(1)$ diminishes at a rate determined by m while n increases. This shows that, for a fixed alphabet size m , the per symbol redundancy $\frac{\hat{R}(I_m^n)}{n}$ diminishes as $n \rightarrow 0$. Therefore, the entropy rate can be achieved for i.i.d. sources with a fixed finite alphabet even when the probability distribution of the source is unknown.

We wish to have the same result for any alphabet however it is not possible. To see its impossibility, let's consider the following toy example. Let $C^n := \{p_k^n : k \in \mathbb{N}\}$ where

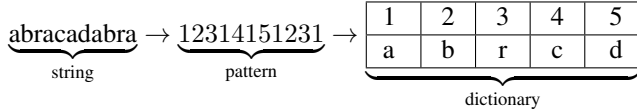
$$p_k^n(\vec{x}) := \begin{cases} 1, & \text{if } \vec{x} = \underbrace{k, k, \dots, k}_{n \text{ many}} \\ 0, & \text{otherwise} \end{cases}$$

Thus, if we know that the distribution of the source is $p_{k_0}^n$ for some $k_0 \in \mathbb{N}$, then we need $H(p_{k_0}^n) = 0$ bits to describe to the sequence.

However, if we only know that the distribution of the source is coming from the class C^n , then we need to know the value of k to describe the sequence. Thus, we need an infinite amount of bits. Hence, the redundancy becomes infinite.

So, even this simple collection C^n has infinite redundancy. However, we can still suggest an approach to some part of the redundancy.

Consider we have a string "abracadabra". We can separate it into 2 parts, namely "pattern" and "dictionary" as follows:



Then, conveying a string is equivalent to conveying its pattern and dictionary. Our goal is to show that patterns can be conveyed efficiently, and the infinite redundancy is actually caused by the problem of conveying the dictionary. We will do this by defining $\hat{R}(I_\Psi^n)$, the pattern redundancy of the collection I^n of all i.i.d. distributions over length n strings drawn from any alphabet:

$$\hat{R}(I_\Psi^n) = \inf_q \sup_{p \in I_\Psi^n} \sup_{\psi \in \Psi^*} \log \left(\frac{p(\psi)}{q(\psi)} \right)$$

and prove that it is bounded via the bounds given in 5.

For instance, let's go back to our toy example. The pattern will always be 1, 1, ..., 1 for the class C^n . Thus, no bit is needed to convey the pattern, thus there is no pattern redundancy. All the redundancy is caused by describing k which is the only element in the dictionary.

Lastly, it is worth mentioning the profiles of patterns. It is defined as the multiplicities of multiplicities of each number of the pattern in the backward order. For instance, the multiplicities of a pattern 12131 are 3, 1, 1, 0, 0 respectively. So, its profile is (0, 0, 1, 0, 2). Profiles are explained in detail in section C of Appendix-A. It is an important concept since every induced i.i.d. distribution assigns the same probability to all patterns of the same profile.

A. Useful results on Patterns

This is the part where we presented the useful results related to patterns given in [2]. These results are essentially important to understand the proofs. Particularly, we evaluated the pattern redundancy for $n = 1$ and $n = 2$, explained the relations between patterns and set partitions, defined profiles, and introduced Shtarkov's sum ([4]) in the Appendix-A of this report.

B. Block Compression

In this section, we will prove, $\forall n \in \mathbb{Z}^+$, that:

$$\frac{3}{2} \log en^{\frac{1}{3}}(1 + o(1)) \leq \hat{R}(I_\Psi^n) \leq \pi \sqrt{\frac{2}{3}} \log e \sqrt{n} \quad (5)$$

It means that the pattern redundancy of memoryless sources grows to infinity as $n \rightarrow \infty$, however, the per-symbol pattern redundancy diminishes as $n \rightarrow \infty$. Its proof is given in [2]. In this part, we will only provide a sketch of the proof for the upper bound on $\hat{R}(I_\Psi^n)$ which is the part that we are mostly interested in as it means getting a diminishing pattern redundancy per symbol asymptotically. Now, let's state the upper bound in 5 as a theorem and give a sketch of its proof:

Theorem 7:

$$\hat{R}(I_\Psi^n) \leq \pi \sqrt{\frac{2}{3}} \log e \sqrt{n}$$

Proof: (sketch of the proof)

- (Step 1): For a collection of distributions $\mathcal{P}$, we will show that $\hat{R}(\mathcal{P}) \leq \log |\mathcal{P}|$. By Shtarkov's sum we have:

$$\hat{R}(\mathcal{P}) = \log \sum_{x \in \mathcal{X}} \max_{p \in \mathcal{P}} p(x) \leq$$

$$\log \sum_{p \in \mathcal{P}} \left(\sum_{x \in \mathcal{X} : \max_{p' \in \mathcal{P}} p'(x) = p(x)} p(x) \right) \leq \log \sum_{p \in \mathcal{P}} 1 = \log |\mathcal{P}|$$

- (Step 2): If $\hat{\mathcal{P}}$ dominates $\mathcal{P}$ (i.e. $\sup_{p \in \hat{\mathcal{P}}} p(x) \geq \sup_{p \in \mathcal{P}} p(x)$), then, by Shtarkov's sum, we have:

$$\hat{R}(\mathcal{P}) \leq \hat{R}(\hat{\mathcal{P}}) \leq \log |\hat{\mathcal{P}}|$$

- (Step 3): We have I_Ψ^n which is the collection of all pattern distributions of memoryless sources. Let's choose one distribution for each profile from this set to form $\hat{I}_\Psi^n$. Then, $|\hat{I}_\Psi^n| = |\Phi^n|$ where Φ^n is the set of profiles of all length n patterns.

Now, observe that every induced i.i.d. distribution assigns the same probability to all patterns of the same profile. Therefore, $\hat{I}_\Psi^n$ dominates I_Ψ^n . Then, by step 2:

$$\hat{R}(I_\Psi^n) \leq \log |\Phi^n|$$

- (Step 4): As we explained in Appendix-A, Φ^n is equivalent to the collection of the set partitions of $[n]$. The size of the collection of the set partitions of $[n]$ is bounded in [13]:

$$|\Phi^n| \leq \exp(\pi \sqrt{\frac{2}{3}} \sqrt{n}) \quad (6)$$

By using the inequality given in 15 together with step 3, we get:

$$\hat{R}(I_\Psi^n) \leq (\pi \sqrt{\frac{2}{3}} \log e) \sqrt{n}$$

C. Sequential Compression

For some applications, we need to encode the symbols as they arrive instead of waiting for the arrival of the whole block.

Let $q(x|\psi_1^n)$ be a probability distribution over $\{1, 2, \dots, \max(\psi_1^n) + 1\}$ representing the probability that the encoder assigns the possible values of ψ_{n+1} after observing ψ_1^n . For example, $q(x|1212)$ is a distribution on $\{1, 2, 3\}$ and $q(x|1234)$ is a distribution on $\{1, 2, 3, 4, 5\}$.

Let q be a sequential encoder. Then, q induces a probability distribution over Ψ^n given by:

$$q(\psi_1^n) = \prod_{i=1}^n q(\psi_i|\psi_1^{i-1}) \quad (7)$$

Two different sequential encoders are proposed in [2], namely $q_{1/2}$ and $q_{2/3}$. $q_{1/2}$ is called the low-redundancy encoder and its pattern redundancy is in order $n^{1/2}$. $q_{2/3}$ is called the low-complexity encoder and its pattern redundancy is in order $n^{2/3}$. Both grow sublinearly with n , but not as slowly as the redundancy in the block compression case. However, as they grow sublinearly with n , we can still say that their per-symbol redundancies (i.e. $\frac{\hat{R}(I_\Psi^n, q_{1/2})}{n}$ and $\frac{\hat{R}(I_\Psi^n, q_{2/3})}{n}$) diminishes as $n \rightarrow \infty$. Now, let's briefly examine these sequential encoders separately.

1) *low-redundancy encoder ($q_{1/2}$)*: Let $\hat{p}_{\psi_1^n}(\psi_1^n) := \sup_{p \in I_\Psi^n} p(\psi_1^n)$ be the maximum-likelihood probability assigned to a pattern $\psi_1^n \in \Psi^n$ by any i.i.d. distribution on I_Ψ^n . Since every i.i.d. distribution assigns the same probability to all patterns of the same profile:

$$\hat{p}_{\psi_1^n}(\psi_1^n) \leq \frac{1}{N(\rho(\psi_1^n))}$$

where $N(\rho(\psi_1^n))$ be the number of patterns with profile $\rho(\psi_1^n)$. Let's define the distribution $\tilde{p}(\psi_1^n)$:

$$\tilde{p}(\psi_1^n) := \frac{1/N(\rho(\psi_1^n))}{\sum_{\psi \in \Psi^n} \frac{1}{N(\rho(\psi))}} = \frac{1}{N(\rho(\psi_1^n))|\Phi^n|}$$

Moreover, for $k \geq n$, let $\Psi^k(\psi_1^n)$ be the set of all patterns that extends ψ_1^n in Ψ^k . Let's define the distribution $p^{\sim k}(\psi_1^n)$:

$$p^{\sim k}(\psi_1^n) := \tilde{p}(\Psi^k(\psi_1^n)) = \sum_{y \in \Psi^k(\psi_1^n)} \tilde{p}(y)$$

Furthermore, let $t_n := 2^{\lceil \log n \rceil}$. Then, we define $q_{1/2}$:

$$q_{1/2}(\psi_n|\psi_1^{n-1}) := \frac{p^{\sim t_n}(\psi_1^n)}{p^{\sim t_n}(\psi_1^{n-1})}$$

Theorem 8: For all $n \in \mathbb{N}$,

$$\hat{R}(I_\Psi^n, q_{1/2}) \leq \frac{4\pi \log e}{\sqrt{3}(2 - \sqrt{2})} \sqrt{n}$$

The sketch of the proof is given in Appendix-C. By theorem 8, we see that the pattern redundancy of the encoder $q_{1/2}$ is indeed in order $n^{1/2}$. Moreover, this redundancy is a little bit higher than what we get in the block compression case.

2) *low-complexity encoder ($q_{2/3}$)*: The evaluation of $q_{1/2}(\psi_1^n)$ is computationally costly, more specifically super-polynomial complexity. Therefore, [2] suggests a low-complexity encoder as well, namely $q_{2/3}$. It has a higher redundancy than $q_{1/2}$ but still its per-symbol pattern redundancy diminishes as $n \rightarrow \infty$.

Moreover, evaluation of $q_{2/3}$ is shown to have linear complexity and its pattern redundancy can be upper bounded by $10n^{2/3}$. However, both the computational complexity and the redundancy analysis contain complex derivations given in [2], therefore we will omit these in this report.

IV. UNIVERSAL COMPRESSION OF MEMORYLESS SOURCES OVER LARGE ALPHABETS

Popular universal coding algorithms, such as Lempel-Ziv algorithm, usually operate on the binary alphabet by converting the source symbols into bits. One reason of this alphabet size reduction is the redundancy of universal encoding. As seen from the equation 4, the redundancy increases with the alphabet size. However, operating on a smaller alphabet, they have the risk of losing the natural correlation between the source symbols. On the other hand, shapes and patterns consider the actual symbol occurring, so they have the potential to lower the redundancy for larger alphabets.

In the section III, we examined the pattern approach and showed that patterns of symbols generated by i.i.d. distributions over any alphabet (including unknown and infinite alphabets) can be compressed with a diminishing per-symbol redundancy. In this section, we will take a broader view of the universal compression of memoryless sources over large alphabets. We will examine 3 approaches, namely the standard compression approach, pattern approach, and shape approach. We will also compare the redundancy rates for these schemes.

A. Standart Compression Approach

Let $\mathcal{A}$ be an m symbol alphabet. If $x \in \mathcal{A}^n$ has l distinct symbols occurring $u_1, u_2, \dots, u_l$ times, then, by typicality:

$$\max_{p \in \Pi_m^n} p(x) = \prod_{j=1}^l \left(\frac{u_j}{n}\right)^{u_j}$$

Then, by Shtarkov's sum, it can be found that:

$$\hat{R}(I_m^n) = \log \left(\sum_{l=1}^{\min\{m,n\}} \binom{m}{l} \sum_{\vec{u} \in \mathcal{U}_l^n} \binom{n}{u_1, \dots, u_l} \prod_{j=1}^l \left(\frac{u_j}{n}\right)^{u_j} \right) \quad (8)$$

Then, using 8, the succession of papers [5], [6], [7], [8], [9], [10], [11], [12] shows:

$$\hat{R}(I_m^n) \leq \frac{m-1}{2} \log \frac{n}{2\pi} + \log \frac{\Gamma(\frac{1}{2})^m}{\Gamma(\frac{m}{2})} + \mathcal{O}\left(\frac{1}{\sqrt{n}}\right) \quad (9)$$

Moreover, they derived equation 4. It is important because it proves that the per-symbol redundancy $\frac{\hat{R}(I_m^n)}{n} \rightarrow 0$ as $n \rightarrow \infty$ for fixed alphabet size m .

Our goal is to analyze at which rate of the alphabet size we will get zero, constant, or infinite per-symbol redundancy. We already have an upper bound for $\hat{R}(I_m^n)$ that is inequality 9. Thus, to achieve our goal, it is useful to derive a lower bound for $\hat{R}(I_m^n)$ as well. Therefore, let's introduce theorem 9:

Theorem 9:

$$\hat{R}(I_m^n) \geq \log\left(\sum_{l=1}^{\min\{m,n\}} \binom{m}{l} \frac{\sqrt{2\pi n}}{e^{l/12}} \binom{n-1}{l-1} \left(\frac{l}{2\pi n}\right)^{l/2}\right)$$

The proof is given in Appendix-C. Now, combining the results of theorem 9 and 9, we can determine at which rate of the alphabet size the per-symbol redundancy will be zero, constant, or infinite:

Theorem 10:

$$\text{If } m = o(n) \implies \hat{R}(I_m^n) \sim \frac{m-1}{2} \log \frac{n}{m}$$

$$\text{If } m = \Theta(n) \implies \hat{R}(I_m^n) = \Theta(n)$$

$$\text{If } n = o(m) \implies \hat{R}(I_m^n) \sim n \log \frac{m}{n}$$

The proof is given in Appendix-C.

B. Pattern Approach

For the pattern approach, [3] presents the same results as in [2] with a few additional results for some special cases. For the sake of brevity, we will not repeat these results and one may refer to section III for the pattern approach.

C. Shape Approach

In contrary to the pattern approach, we have an ordered alphabet this time. Let's say we have the roman alphabet with the alphabetic order. Consider we have a string "abracadabra" over this alphabet. We can separate it into 2 parts, namely "shape" and "set of symbols" as follows:

$$\underbrace{\text{abracadabra}}_{\text{string}} \rightarrow \underbrace{12513141251}_{\text{shape}} \rightarrow \underbrace{\begin{array}{|c|c|c|c|c|} \hline 1 & 2 & 3 & 4 & 5 \\ \hline a & b & c & d & r \\ \hline \end{array}}_{\text{set of symbols}}$$

Then, conveying a string is equivalent to conveying its shape and its set of symbols. Our goal is to show that shapes can be conveyed somewhat efficiently, and the infinite redundancy is actually caused by the problem of conveying the set of symbols.

We will define $\hat{R}(I_S^n)$, the shape redundancy of the collection I^n of all i.i.d. distributions over length n strings drawn from any finite or infinite alphabet as:

$$\hat{R}(I_S^n) = \inf_q \sup_{p \in I_S^n} \sup_{s \in S^*} \log \frac{p(s)}{q(s)}$$

Unlike the diminishing per-symbol pattern redundancy or possibly diminishing, constant, or infinite standard per-symbol

redundancy, we will show that the shape redundancy per-symbol is between 0.027 and 1. Formally, we want to show:

$$0.027 < \frac{1}{4} \log\left(\frac{4}{\pi e^{1/6}}\right) \leq \frac{\hat{R}(I_S^n)}{n} \leq 1 \quad (10)$$

We will show the inequality in 10 by proving the upper bound and the lower bound separately for a block compression scheme. In [3], they also suggest an approach for sequential compression, but they do not present any result on the redundancy. We will revisit this issue in section V.

Lastly, it is worth mentioning the types of shapes. It is defined as the multiplicities of each number of the shape. For instance, the type of a shape 12513141251 is (5,2,1,1,2). Types are explained in detail in section B of Appendix-B. It is an important concept since every induced i.i.d. distribution assigns the same probability to all shapes of the same type.

1) *Useful Results on Shapes:* This is the part where we presented some results related to shapes given in [3]. These results are especially useful to understand the proofs. Particularly, we estimated shape redundancy for short strings, we explained the classification of shapes, and the relation between the shapes and the ordered set partitions in the Appendix-B of this report.

2) *Upper Bound given in the inequality 10:*

Theorem 11: For all n ,

$$\hat{R}(I_S^n) \leq n - 1$$

Proof: (Analogous to the proof of theorem 7)

As we showed in the proof of theorem 7, for a collection of distributions $\mathcal{P}$, we have $\hat{R}(\mathcal{P}) \leq \log |\mathcal{P}|$.

Moreover, if $\hat{\mathcal{P}}$ dominates $\mathcal{P}$ (i.e. $\sup_{p \in \mathcal{P}} p(x) \geq \sup_{p \in \hat{\mathcal{P}}} p(x)$), then:

$$\hat{R}(\mathcal{P}) \leq \hat{R}(\hat{\mathcal{P}}) \leq \log |\hat{\mathcal{P}}|$$

Let $u \in \mathcal{U}^n$ be a type and $\mathcal{S}_u$ be the set of shapes of type u . Let's define:

$$\hat{I}_S^n := \{\bar{p}_u : u \in \mathcal{U}^n\}$$

where $\bar{p}_u$ is the uniform distribution assigning probability $\frac{1}{|\mathcal{S}_u|}$ to every shape in $\mathcal{S}_u$.

Now, observe that every induced i.i.d. distribution assigns the same probability to all shapes of the same type. Thus, I_S^n induces the same probability on all shapes in $\mathcal{S}_u$. Therefore, the probability it assigns to every shape can't exceed $\frac{1}{|\mathcal{S}_u|}$. Hence, $\hat{I}_S^n$ dominates I_S^n . Then, by step 2 and using the definition of $\hat{I}_S^n$:

$$\hat{R}(I_S^n) \leq \log |\hat{I}_S^n| = \log |\mathcal{U}^n|$$

Finally, using the identity in 12, the proof follows ■

As a result of this theorem, we have $\frac{\hat{R}(I_S^n)}{n} \leq 1$ for all n . Therefore, per-symbol shape redundancy is less than 1.

3) *Lower Bound given in the inequality 10:*

Theorem 12: For all $n > 1$,

$$\hat{R}(I_S^n) \geq \frac{n}{4} \log \frac{4}{\pi e^{1/6}} > 0.027n$$

The proof is given in the Appendix-C. As a result of this theorem, we have $\frac{\hat{R}(I_S^n)}{n} \geq 0.027$ for all $n > 1$. Therefore, per-symbol shape redundancy is at least 0.027.

D. Comparison

In [3], 3 approaches for universal compression of memoryless sources over large alphabets are examined. In the standard compression approach, we determined at which rate of the alphabet size m (according to the number of symbols n) the per-symbol redundancy can be limited. We saw that we can have a finite or diminishing redundancy unless $n = o(m)$, which means that our alphabet is so big that the number of symbols is insignificant compared to the alphabet size. However, in this case, we suggest that using the pattern and shape approaches, we can limit some parts of the redundancy. We proved that conveying the pattern of a string can be done with a diminishing redundancy per symbol as n grows. Also, we proved that conveying the shape of a string can be done with a per-symbol redundancy between 0.027 and 1 as n grows. Thus, one may think that the patterns are more useful than shapes since pattern redundancy per symbol diminishes. However, we still need to convey the dictionary on top of conveying a pattern although it is sufficient to convey the set of symbols on top of conveying a shape. Clearly, conveying the dictionary is at least as hard as conveying the set of symbols. This makes the shape approach more advantageous than the pattern approach in some cases. However, the pattern approach is proven to work for both block and sequential compression schemes, although the shape approach and the standard approach are designed for block compression only.

V. RESEARCH PROPOSAL

In this report, we examined lots of techniques for the various cases of universal compression problems. However, all the bounds on the redundancy or the error probability they provide are only reasonable as the number of symbols grows. Thus, they do not promise anything about the one-shot applications. Therefore, there might still be room for improvement in these variations of universal coding problems for the one-shot case.

Moreover, the discussed shape ideas help us for the case when the alphabet is ordered and the discussed pattern ideas are helpful for the case when there is no order in the alphabet. However, there might be some kind of partial order in the alphabet, and they do not suggest anything on how to use this information.

Lastly, in [3] section VI-E, they suggest an idea about a sequential compression scheme, but do not present any result on its redundancy.

In this section, we will propose our ideas about how to improve these issues.

A. One-shot vs Asymptotical

I have worked on Strong Functional Representation Lemma (SFRL, [14]) and Poisson Matching Lemma (PML, [15]) in my semester projects. They both have some one-shot applications for various problems in information theory. In [16], we presented a universal version of SFRL and we discussed our ideas about using these techniques for the universal compression problem with side information. It is still an ongoing research now. In the future, we may study the other variations of the universal compression problems in this

functional representation perspective by utilizing the pattern and shape ideas.

B. Extension of ideas for partially ordered alphabets

In some applications, we might have some kind of partial ordering in the alphabet. For instance, consider:

$$\mathcal{A} = \{a, b, \dots, z, A, B, \dots, Z\}$$

to be an alphabet of size 52 where there is an order between the small letters (i.e. $a < b < \dots < z$) and there is an order between capital letters (i.e. $A < B < \dots < Z$). But, there is no order between a capital and a small letter.

In this case, our goal is to characterize a property (such as pattern and shape) for the strings, namely weak-shape. Using this information about the partial order, we want to define and analyze the weak-shape redundancy as [2] and [3] did for the pattern and shape.

C. Sequential Compression of Shapes

In VI-E of [3], they define some concept of sequential shape to be able to describe the shape of incoming sequence without waiting for the arrival of the whole block. They stated that the sequential shape of every prefix of a string is the corresponding prefix of its sequential shape. However, they do not present any result on the redundancy. Therefore, it might be a nice future direction to consider the sequential shape redundancy and how to bound it.

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APPENDIX A USEFUL RESULTS ON PATTERNS

Let $\Psi(\vec{x})$ be the pattern of a string $\vec{x} \in \mathcal{A}^n$ where $\mathcal{A}$ is the alphabet.

$$\implies \Psi(\text{abracadabra}) = 12314151231$$

Let Ψ^n be the set of all possible patterns of a string of length n .

$$\implies \Psi^1 = \{1\}, \Psi^2 = \{11, 12\}, \Psi^3 = \{111, 112, 121, 122, 123\}$$

Let Ψ^* be the set of all possible patterns of a string of any length.

$$\implies \Psi^* = \{1, 11, 12, 111, 112, 121, 122, 123, \dots\}$$

Then, notice that every probability distribution p over $\mathcal{A}^*$ (the set of strings of any length) induces a probability distribution $p_\Psi(\psi)$ for a pattern ψ over Ψ^* :

$$p_\Psi(\psi) := p(x \in \mathcal{A}^* : \Psi(x) = \psi)$$

Moreover, for a collection of distributions $\mathcal{P}$ over $\mathcal{A}^*$, let $\mathcal{P}_\Psi := \{p_\Psi : p \in \mathcal{P}\}$. Then, we define the worst-case pattern redundancy of $\mathcal{P}$ as the redundancy of conveying the pattern of a string. Then:

$$\hat{R}(\mathcal{P}_\Psi) = \inf_q \sup_{p \in \mathcal{P}_\Psi} \sup_{\psi \in \Psi^*} \log\left(\frac{p(\psi)}{q(\psi)}\right) \quad (11)$$

Here, ψ is selected over Ψ^* as the worst-case pattern. And, p is selected over $\mathcal{P}_\Psi$ as the worst-case probability distributions by which patterns are generated. Lastly, q is the best choice of the probability distribution that induces the length of the codewords to describe each pattern.

Let I^n be the collection of an arbitrary i.i.d. distribution over length n strings. Our goal is to prove $\frac{\hat{R}(I_\Psi^n)}{n} \rightarrow 0$ as $n \rightarrow \infty$. And, we present both a block compression scheme and a sequential compression scheme achieving it in section III-B and section III-C respectively.

Keeping this goal in mind, let's present some useful results on patterns discussed in [2]:

A. The Pattern Redundancy of I_1 and I_2

For small values of n , particularly for $n = 1$ and $n = 2$, we can actually calculate $\hat{R}(I_\Psi^n)$ using Shtarkov's sum. By Shtarkov's sum and equation 11, we can calculate the worst-case pattern redundancy of the class I_Ψ^n as:

$$\hat{R}(I_\Psi^n) = \log\left(\sum_{\rho \in \Phi^n} \sum_{\psi \in \Psi_\rho} \sup_{p \in I_\Psi^n} p(\psi)\right)$$

For $n = 1$, using $\hat{R}(I_\Psi^1) = \log(1) = 0$ as there is only 1 pattern in I_Ψ^1 .

For $n = 2$, $\Psi^2 = \{11, 12\}$. Then;

$$\hat{R}(I_\Psi^2) = \log\left(\sup_{p \in I_\Psi^2} p(11) + \sup_{p \in I_\Psi^2} p(12)\right) = \log(1 + 1) = 1$$

since $p(11)$ becomes 1 for any constant i.i.d. distribution and $p(12)$ becomes 1 for an i.i.d. continuous uniform distribution on $[0, 1]$.

B. Set partitions and Patterns

Let $[n] := \{1, 2, \dots, n\}$ and $[0] := \emptyset$. Let $\mathcal{S}^n$ be the set of all partitions of $[n]$ and $\mathcal{S}^* := \bigcup_{n=0}^{\infty} \mathcal{S}^n$. For instance:

$$\mathcal{S}^0 = \{\emptyset\}, \mathcal{S}^1 = \{\{\{1\}\}\}, \mathcal{S}^2 = \{\{\{1, 2\}\}, \{\{1\}, \{2\}\}\}$$

Theorem 13: The function $f_\Psi : \Psi^* \rightarrow \mathcal{S}^*$ defined (for $\psi = \psi_1 \dots \psi_{|\psi|}$) as:

$$f_\Psi(\psi) := \{\{i : \psi_i = j\} : 1 \leq j \leq \max_{1 \leq i \leq |\psi|} \psi_i\}$$

is a bijection. Furthermore, for every n :

$$f_\Psi(\Psi^n) = \mathcal{S}^n$$

This theorem means that set partitions are equivalent to patterns. For example, the pattern $\psi = 12131$, is equivalent to the set partition given by $f_\Psi(\psi)$ which equals:

$$\{\{i : \psi_i = 1\}, \{i : \psi_i = 2\}, \{i : \psi_i = 3\}\} = \{\{1, 3, 5\}, \{2\}, \{4\}\}$$

C. Profiles

We define multiplicities of a pattern ψ as $\mu_k := |\{\psi_i : \psi_i = k\}|, \forall k \in \mathbb{Z}^+$. So, for a pattern $\psi = 12131$, we have:

$$\psi = 12131 \implies \mu_1 = 3, \mu_2 = 1, \mu_3 = 1, \mu_k = 0 \text{ for } k > 3$$

Additionally, we define the prevalences of the multiplicities of a pattern ψ as $\rho_k := |\{\mu_l : \mu_l = k\}|, \forall 1 \leq k \leq |\psi|$. So, for a pattern $\psi = 12131$, we have:

$$\psi = 12131 \implies \rho_1 = 2, \rho_2 = 0, \rho_3 = 1, \rho_4 = 0, \rho_5 = 0$$

Finally, we can define the profile of a pattern as its prevalences in the backward order. So, for a pattern $\psi = 12131$, we have:

$$\psi = 12131 \implies \rho(\psi) = (0, 0, 1, 0, 2)$$

Note that, for any profile $\rho \in \Phi^n$, we have $\sum_{k=1}^n k \rho_k = n$.

Moreover, the profile of a set partition of the set $[n]$ is defined as prevalences in the backward order as well with ρ_k as the number of elements of the sets, with size k , in the partition. For example:

$$S = \{\{1, 3, 5\}, \{2\}, \{4\}\} \implies \rho(S) = (0, 0, 1, 0, 2)$$

Then, there is an equivalency between the set of all profiles of all length n patterns (that is Φ^n) and the set of profiles of the all set partitions of $[n]$. Moreover, we have:

$$f_\Psi(\Psi_\rho) = S_\rho$$

where Ψ_ρ is the collection of all patterns of profile ρ and S_ρ is the collection of all partitions with profile ρ .

Note that profile is a very important concept since every induced i.i.d. distribution assigns the same probability to all patterns of the same profile.

D. Shtarkov's Sum

In [4], Shtarkov finds the distribution q that achieves the worst-case redundancy $\hat{R}(\mathcal{P})$ over a class $\mathcal{P}$ (given in equation 3) as:

$$q^*(x) = \frac{\sup_{p \in \mathcal{P}} p(x)}{\sum_{x' \in \mathcal{X}} \sup_{p \in \mathcal{P}} p(x')}$$

Then, the redundancy of a collection $\mathcal{P}$ of distributions over $\mathcal{X}$ is:

$$\hat{R}(\mathcal{P}) = \log\left(\sum_{x \in \mathcal{X}} \sup_{p \in \mathcal{P}} p(x)\right)$$

So, for instance, for our toy example, $\hat{R}(\mathcal{P})$ becomes $\log(\sum_{k \in \mathbb{N}} 1)$ which is infinite as expected.

APPENDIX B USEFUL RESULTS ON SHAPES

Let $\mathcal{A}$ be a possibly infinite (countable or uncountable) alphabet with order. Let $x = x_1 x_2 \dots x_n \in \mathcal{A}^n$ be a string. We show the rank of x_i in x with $\rho_x(x_i)$. For example, if $x = \text{abracadabra}$:

$$\rho_x(a) = 1, \rho_x(b) = 2, \rho_x(c) = 3, \rho_x(d) = 4, \rho_x(r) = 5$$

Then, we show the shape of x with $\mathcal{S}(x)$ as:

$$\mathcal{S}(x) = \mathcal{S}(\text{abracadabra}) = 12513141251$$

Then, let $\mathcal{S}(\mathcal{A}^n)$ be the set of all strings in $\mathcal{A}^n$. Then, for $\mathcal{B} = \{0, 1\}$:

$$\mathcal{S}(\mathcal{B}) = \{1\}, \mathcal{S}(\mathcal{B}^2) = \{11, 12, 21\}$$

$$\mathcal{S}(\mathcal{B}^3) = \{111, 112, 221, 121, 122, 211, 212\}$$

Let $\mathcal{S}^n$ be the set of all length n shapes. Then:

$$\mathcal{S}^1 = \{1\}, \mathcal{S}^2 = \{11, 12, 21\}$$

$$\mathcal{S}^3 = \{111, 112, 221, 121, 122, 211, 212, 123, 132, 213, 231, 312, 321\}$$

Then, observe that a string is a shape if and only if it contains $i - 1$ for every positive integer $i \geq 2$ it contains.

Every probability distribution p over $\mathcal{A}^*$ induces a probability distribution p_S over $\mathcal{S}^*$ where:

$$p_S(s) := p(\{x \in \mathcal{A}^* : \mathcal{S}(x) = s\})$$

For instance, let p be a uniform distribution over $\{a, b\}^2$ with order $a < b$. Then:

$$p(11) = p(\{aa, bb\}) = \frac{1}{2}$$

$$p(12) = p(\{ab\}) = \frac{1}{4}, p(21) = p(\{ba\}) = \frac{1}{4}$$

Then, for a collection of distributions $\mathcal{P}$ over $\mathcal{A}^*$ (or over any subset of $\mathcal{A}^*$), we define:

$$\mathcal{P}_S := \{p_S : p \in \mathcal{P}\}$$

Similarly to the patterns, we define the worst-case shape redundancy of $\mathcal{P}$ as the redundancy of conveying the pattern of a string. Then:

$$\hat{R}(\mathcal{P}_S) = \inf_q \sup_{p \in \mathcal{P}_S} \sup_{s \in \mathcal{S}^*} \log \frac{p(s)}{q(s)}$$

Now, let's proceed with some more specific results in the following subsections:

A. The Shape Redundancy of I_1 and I_2

There is only 1 distribution on $\mathcal{S}^1$, therefore $\hat{R}(I_S^1) = 0$.

Now, consider $\hat{R}(I_S^2)$ which is the shape redundancy of I_S^2 which is the collection of i.i.d. induced distributions over $\mathcal{S}^2 = \{11, 12, 21\}$. By Shtarkov's sum:

$$\begin{aligned} \hat{R}(I_S^2) &= \log\left(\sum_{s \in \mathcal{S}^2} \sup_{p \in I_S^2} p(s)\right) \\ &= \log\left(\sup_{p \in I_S^2} p(11) + \sup_{p \in I_S^2} p(12) + \sup_{p \in I_S^2} p(21)\right) \\ &= \log\left(1 + \frac{1}{2} + \frac{1}{2}\right) = \log(2) = 1 \end{aligned}$$

since $p(11)$ becomes 1 for any constant i.i.d. distribution and $p(12)$ and $p(21)$ becomes $\frac{1}{2}$ for an i.i.d. continuous uniform distribution on $[0, 1]$.

B. Classification of Shapes

To analyze $\hat{R}(I_S^n)$, we classify $\mathcal{S}^n$ into the sets of equiprobable shapes. Let $\mathcal{S}_m^n$ be the set of length n shapes with m symbols. For example:

$$\begin{aligned}\mathcal{S}_1^3 &= \{111\}, \mathcal{S}_2^3 = \{112, 211, 121, 212, 211, 122\} \\ \mathcal{S}_3^3 &= \{123, 132, 213, 231, 312, 321\}\end{aligned}$$

Moreover, for all $1 \leq i \leq m \leq n$, let μ_i be the multiplicity of i in $\mathcal{S}_m^n$. For example, for shape 12131:

$$\mu_1 = 3, \mu_2 = 1, \mu_3 = 1$$

And, let's define the type of a shape as $u(s) := (\mu_1, \mu_2, \dots, \mu_m)$. For example:

$$u(12131) = (3, 1, 1)$$

Note that this concept is very important since every induced i.i.d. distribution assigns the same probability to all shapes of the same type.

C. Ordered partitions of positive integers and shapes

An ordered partition of a positive integer n is a vector of positive integers summing to n , which we call "parts". For example, (1,2,1) is an ordered partition of 4 into 3 parts.

Let $\mathcal{U}_m^n$ be the set of all ordered partitions of n into m parts and $\mathcal{U}_n = \bigcup_{m=1}^n \mathcal{U}_m^n$. For example:

$$\mathcal{U}_1^4 = \{(4)\}, \mathcal{U}_3^4 = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$$

$$\mathcal{U}^4 = \{(4), (1, 3), (3, 1), (1, 1, 2), (1, 2, 1), (2, 1, 1), (1, 1, 1, 1)\}$$

Then, we have:

$$|\mathcal{U}_m^n| = \binom{n-1}{m-1} \text{ and } |\mathcal{U}^n| = 2^{n-1} \quad (12)$$

Now, observe that the type of any length n shape is equivalent to an ordered partition of n . Then, we have:

$$|\mathcal{S}_m^n| = \binom{n-1}{m-1} \text{ and } |\mathcal{S}^n| = 2^{n-1} \quad (13)$$

Moreover, for $u = u_1 u_2 \dots u_m \in \mathcal{U}_m^n$, let $\mathcal{S}_u$ be the collection of shapes of type u . Then:

$$|\mathcal{S}_u| = \binom{n}{u_1, u_2, \dots, u_m} \quad (14)$$

D. Shapes and Ordered Set Partitions

An ordered partition of a set S into m parts is an ordered m -tuple of disjoint non-empty subsets of S whose union is S . And, its type is the cardinalities of these subsets. For example, $(\{1, 3, 5\}, \{2\}, \{4\})$ is an ordered set partition of $\{1, 2, 3, 4, 5\}$ with type (3, 1, 1).

Moreover, let $\mathcal{T}^n$ be the set of all ordered partitions of $[n]$. Let $\mathcal{T}_m^n$ be the set of all ordered partitions of $[n]$ into m parts. And, let $\mathcal{T}_u$ be the set of all ordered partitions of type u .

Similarly to the patterns, we will define a mapping f_S from the set of all shapes to the set of all ordered partitions. For example, for the shape 12131, we define:

$$\begin{aligned}f_S(12131) &= (\{i : s_i = 1\}, \{i : s_i = 2\}, \{i : s_i = 3\}) \\ &= (\{1, 3, 5\}, \{2\}, \{4\})\end{aligned}$$

Theorem 14: The function f_S is a bijection. Moreover:

$$\begin{aligned}\forall n \geq 0 &\implies f_S(\mathcal{S}^n) = \mathcal{T}^n \\ \forall n \geq m \geq 0 &\implies f_S(\mathcal{S}_m^n) = \mathcal{T}_m^n \\ \forall \text{ types } u &\implies f_S(\mathcal{S}_u) = \mathcal{T}_u\end{aligned}$$

APPENDIX C

PROOF OF SOME THEOREMS

Theorem 1: For all $\rho \in [0, 1]$, and $M = \lceil 2^{NR} \rceil$

$$P_e(y) \leq M^{-\rho} \left[\sum_{\vec{x}_N} Q_N(\vec{x}|y)^{\frac{1}{1+\rho}} \right]^{1+\rho}$$

Proof: A source sequence $\vec{x}$ is decoded incorrectly for an encoder f if f maps some more likely $\vec{x}'$ into the same codeword. Thus:

$$P_e(y) \leq \sum_{\vec{x}} Q_N(\vec{x}|y) P_N \left[\bigcup_{\vec{x}' \neq \vec{x}: C} \{f : f(\vec{x}') = f(\vec{x})\} \right]$$

where C represents the condition $\{Q_N(\vec{x}'|y) \geq Q_N(\vec{x}|y)\}$. Now by using the following identities:

$$P_N \left[\bigcup_i A_i \right] \leq \left[\sum_i P_N[A_i] \right]^\rho, \forall \{A_i\} \text{ and } \forall \rho \in [0, 1]$$

$$P_N[\{f : f(\vec{x}') = f(\vec{x})\}] = \frac{1}{M}$$

Note that the 2nd one is coming from the definition of random ensembles. So, we will get the following by using them together with the previous inequality for $P_e(y)$:

$$P_e(y) \leq \sum_{\vec{x}} Q_N(\vec{x}|y) \left[\sum_{\vec{x}' \neq \vec{x}: C} \frac{1}{M} \right]^\rho$$

Then, $\forall s \geq 0$, we get:

$$P_e(y) \leq \sum_{\vec{x}} Q_N(\vec{x}|y) \left[\frac{1}{M} \sum_{\vec{x}'} \left(\frac{Q_N(\vec{x}'|y)}{Q_N(\vec{x}|y)} \right)^s \right]^\rho$$

Finally, choosing $s = \frac{1}{1+\rho}$, we complete the proof. $\blacksquare$

Theorem 5: Given any (N, R) , and a source with side information with the probability assignment $Q_N(\vec{x}|y)$, there exists an (N, R) block encoder mapping with an error probability for each y satisfying:

$$P_e(y) \leq \min_{0 \leq \rho \leq 1} \max_P 2^{\Gamma_N} \left[\sum_{\vec{x}} \sum_y P(y) Q_N(\vec{x}|y)^{\frac{1}{1+\rho}} \right]^{1+\rho}$$

where $\Gamma_N = -\rho(NR - \log\lceil\log|\mathcal{Y}|\rceil) + 1$

Proof:

The decoder will assume $P^*(y)$, but let's use a uniform distribution on y at the encoder. Then, by the inequality in 1, there exists a code, $code_1$, whose error probability satisfying:

$$P_{e1}(P^*) \leq 2^{-NR\rho} \left[\sum_{\vec{x}} w_P(\vec{x})^{\frac{1}{1+\rho}} \right]^{1+\rho}$$

Then, $P_{e1}(y, P^*) \leq 2 \times 2^{-NR\rho} \left[\sum_{\vec{x}} w_P(\vec{x})^{\frac{1}{1+\rho}} \right]^{1+\rho}$ for at least half of y values.

Similarly, assigning 0 probability on those y 's and assuming a uniform distributions on the remaining y 's, we can show that $\exists code_2$ such that $P_{e2}(y, P^*) \leq 2 \times 2^{-NR\rho} \left[\sum_{\vec{x}} w_P(\vec{x})^{\frac{1}{1+\rho}} \right]^{1+\rho}$ for at least half of the remaining half of the y values.

Now, let's choose $m \in \mathbb{N}$ such that $m \leq \lceil \log|\mathcal{Y}| \rceil \leq m + 1$. Now, choose encoder as it uses the first one of $code_1, code_2, \dots, code_m$ which will decode the input $\vec{x}$ correctly. Namely:

$$\text{enc}(\vec{x}) = code_k(\vec{x}) \text{ where } k = \min_{1 \leq i \leq m} \{i : \text{dec}(code_i(\vec{x})) = \vec{x}\}$$

This way the code will be decoded correctly whenever any of the $code_1, code_2, \dots, code_m$ decoded correctly, and therefore the error probability bounds that we found for some specific sets of y values are valid for every y . Here, we need $\log m$ bits to describe the specific set of y values which satisfy $P_{ek}(y, P^*) \leq 2 \times 2^{-NR\rho} \left[\sum_{\vec{x}} w_P(\vec{x})^{\frac{1}{1+\rho}} \right]^{1+\rho}$ and we need NR bits to describe $code_k(\vec{x})$. Therefore, the actual rate of our code will be:

$$R' = R + \frac{\log m}{N} \leq R + \frac{\log(\lceil \log|\mathcal{Y}| \rceil)}{N}$$

Therefore, the proof follows. ■

Theorem 7:

$$\hat{R}(I_\Psi^n) \leq \pi \sqrt{\frac{2}{3}} \log e \sqrt{n}$$

Proof: (sketch of the proof)

- (Step 1): For a collection of distributions $\mathcal{P}$, we will show that $\hat{R}(\mathcal{P}) \leq \log |\mathcal{P}|$.

By Shtarkov's sum we have:

$$\hat{R}(\mathcal{P}) = \log \sum_{x \in \mathcal{X}} \max_{p \in \mathcal{P}} p(x) \leq$$

$$\log \sum_{p \in \mathcal{P}} \left(\sum_{x \in \mathcal{X} : \max_{p' \in \mathcal{P}} p'(x) = p(x)} p(x) \right) \leq \log \sum_{p \in \mathcal{P}} 1 = \log |\mathcal{P}|$$

- (Step 2): If $\hat{\mathcal{P}}$ dominates $\mathcal{P}$ (i.e. $\sup_{p \in \hat{\mathcal{P}}} p(x) \geq \sup_{p \in \mathcal{P}} p(x)$), then, by Shtarkov's sum, we have:

$$\hat{R}(\mathcal{P}) \leq \hat{R}(\hat{\mathcal{P}}) \leq \log |\hat{\mathcal{P}}|$$

- (Step 3): We have I_Ψ^n which is the collection of all pattern distributions of memoryless sources. Let's choose one distribution for each profile from this set to form $\hat{I}_\Psi^n$.

Then, $|\hat{I}_\Psi^n| = |\Phi^n|$ where Φ^n is the set of profiles of all length n patterns.

Now, observe that every induced i.i.d. distribution assigns the same probability to all patterns of the same profile. Therefore, $\hat{I}_\Psi^n$ dominates I_Ψ^n . Then, by step 2:

$$\hat{R}(I_\Psi^n) \leq \log |\Phi^n|$$

- (Step 4): As we explained, Φ^n is equivalent to the collection of the set partitions of $[n]$. The size of the collection of the set partitions of $[n]$ is bounded in [13]:

$$|\Phi^n| \leq \exp(\pi \sqrt{\frac{2}{3}} \sqrt{n}) \quad (15)$$

By using the inequality given in 15 together with step 3, we get:

$$\hat{R}(I_\Psi^n) \leq (\pi \sqrt{\frac{2}{3}} \log e) \sqrt{n}$$

Theorem 8: For all $n \in \mathbb{N}$,

$$\hat{R}(I_\Psi^n, q_{1/2}) \leq \frac{4\pi \log e}{\sqrt{3}(2 - \sqrt{2})} \sqrt{n}$$

Proof: (sketch of the proof)

$$\hat{R}(I_\Psi^n, q_{1/2}) = \max_{\psi_1^n} \log \frac{\hat{p}_{\psi_1^n}(\psi_1^n)}{q_{1/2}(\psi_1^n)}$$

is the worst-case regret of using $q_{1/2}$ instead of $\hat{p}_{\psi_1^n}$.

$$\Rightarrow \frac{\hat{p}_{\psi_1^n}(\psi_1^n)}{q_{1/2}(\psi_1^n)} = \frac{\hat{p}_{\psi_1^n}(\psi_1^n)}{p^{\sim t_n}(\psi_1^n)} \frac{p^{\sim t_n}(\psi_1^n)}{q_{1/2}(\psi_1^n)}$$

The proof follows by upper bounding these fractions separately using lemma 16 and lemma 17 from [2]. ■

Theorem 9:

$$\hat{R}(I_m^n) \geq \log \left(\sum_{l=1}^{\min\{m,n\}} \binom{m}{l} \frac{\sqrt{2\pi n}}{e^{l/12}} \binom{n-1}{l-1} \left(\frac{l}{2\pi n}\right)^{l/2} \right)$$

Proof: Using Feller's bound given in [17] with some manipulations, we can lower bound the right-hand side of 8 to get:

$$\begin{aligned} \hat{R}(I_m^n) &\geq \log \left(\sum_{l=1}^{\min\{m,n\}} \binom{m}{l} \frac{1}{e^{l/12}} \sum_{\vec{u} \in \mathcal{U}_l^n} \sqrt{\frac{2\pi n}{(n/l)^l}} \left(\frac{1}{2\pi}\right)^{l/2} \right) \\ &= \log \left(\sum_{l=1}^{\min\{m,n\}} \binom{m}{l} \frac{\sqrt{2\pi n}}{e^{l/12}} \binom{n-1}{l-1} \left(\frac{l}{2\pi n}\right)^{l/2} \right) \end{aligned}$$

where the last equality follows from equation 12. ■

Theorem 10:

$$\text{If } m = o(n) \Rightarrow \hat{R}(I_m^n) \sim \frac{m-1}{2} \log \frac{n}{m}$$

$$\text{If } m = \Theta(n) \Rightarrow \hat{R}(I_m^n) = \Theta(n)$$

$$\text{If } n = o(m) \implies \hat{R}(I_m^n) \sim n \log \frac{m}{n}$$

Proof: (sketch of the proof)

$m = o(n)$ case: The upper bound for $\hat{R}(I_m^n)$ follows from the inequality in 9 because the right-hand side of 9 is $\sim \frac{m-1}{2} \log \frac{n}{m}$. And, the lower bound follows from theorem 9 by retaining only the $l = m$ term from the expression on the right-hand side.

$m = \Theta(n)$ case: Again, the upper bound for $\hat{R}(I_m^n)$ follows from the inequality in 9. And, the lower bound follows from theorem 9 by retaining only the $l = \lceil \frac{nm}{n+m+1} \rceil$ term followed by a few manipulations.

$n = o(m)$ case: The upper bound for $\hat{R}(I_m^n)$ follows from 9 by choosing $l = \min\{m, n\}$ and applying Feller's bound (Corrolaary 1 in [3]). And, the lower bound follows from theorem 9 by retaining only the $l = n$ term from the expression on the right-hand side. ■

Theorem 11: For all n ,

$$\hat{R}(I_S^n) \leq n - 1$$

Proof: (Analogous to the proof of theorem 7)

As we showed in the proof of theorem 7, for a collection of distributions $\mathcal{P}$, we have $\hat{R}(\mathcal{P}) \leq \log |\mathcal{P}|$.

Moreover, if $\hat{\mathcal{P}}$ dominates $\mathcal{P}$ (i.e. $\sup_{p \in \hat{\mathcal{P}}} p(x) \geq \sup_{p \in \mathcal{P}} p(x)$), then:

$$\hat{R}(\mathcal{P}) \leq \hat{R}(\hat{\mathcal{P}}) \leq \log |\hat{\mathcal{P}}|$$

Let $u \in \mathcal{U}^n$ be a type and $\mathcal{S}_u$ be the set of shapes of type u . Let's define:

$$\hat{I}_S^n := \{\bar{p}_u : u \in \mathcal{U}^n\}$$

where $\bar{p}_u$ is the uniform distribution assigning probability $\frac{1}{|\mathcal{S}_u|}$ to every shape in $\mathcal{S}_u$.

Now, observe that every induced i.i.d. distribution assigns the same probability to all shapes of the same type. Thus, I_S^n induces the same probability on all shapes in $\mathcal{S}_u$. Therefore, the probability it assigns to every shape can't exceed $\frac{1}{|\mathcal{S}_u|}$. Hence, $\hat{I}_S^n$ dominates I_S^n . Then, by step 2 and using the definition of $\hat{I}_S^n$:

$$\hat{R}(I_S^n) \leq \log |\hat{I}_S^n| = \log |\mathcal{U}^n|$$

Finally, using the identity in 12, the proof follows ■

Theorem 12: For all $n > 1$,

$$\hat{R}(I_S^n) \geq \frac{n}{4} \log \frac{4}{\pi e^{1/6}} > 0.027n$$

Proof: For any $s \in \mathcal{S}^*$, let

$$\mathcal{S}_p^{-1}(s) = \{x \in \mathcal{A} : \mathcal{S}(x) = s \text{ and } p(x) > 0\}$$

be the support of a shape s with respect to the distribution p . Then, for all $s \in \mathcal{S}^n$:

$$\sup_{p \in I_S^n} p(s) = \sup_{p \in I^n} p(\mathcal{S}_p^{-1}(s)) \geq \sup_{p \in I^n} \max_{x \in \mathcal{S}_p^{-1}(s)} p(x) = \Pi_{j=1}^m \left(\frac{u_j}{n}\right)^{u_j}$$

where the last equality follows from typicality argument assuming the type of s is $u = (u_1, u_2, \dots, u_m)$.

Then, using this lower-bound in the Shtarkov's sum formula:

$$\begin{aligned} \hat{R}(I_S^n) &= \log \left(\sum_{m=1}^n \sum_{u \in \mathcal{U}_m^n} \sum_{s \in \mathcal{S}_u} \sup_{p \in I_S^n} p(s) \right) \\ &\geq \log \left(\sum_{m=1}^n \sum_{u \in \mathcal{U}_m^n} \sum_{s \in \mathcal{S}_u} \Pi_{j=1}^m \left(\frac{u_j}{n}\right)^{u_j} \right) \\ &\stackrel{(a)}{=} \log \left(\sum_{m=1}^n \sum_{u \in \mathcal{U}_m^n} \binom{n}{u_1, u_2, \dots, u_m} \Pi_{j=1}^m \left(\frac{u_j}{n}\right)^{u_j} \right) \\ &\stackrel{(b)}{\geq} \log \left(\sum_{m=1}^n \frac{1}{e^{m/12}} \sum_{u \in \mathcal{U}_m^n} \sqrt{\frac{2\pi n}{(n/m)^m}} \left(\frac{1}{\sqrt{2\pi}}\right)^m \right) \\ &\stackrel{(c)}{\geq} \log \left(\sum_{m=1}^n \frac{1}{e^{m/12}} \binom{n-1}{m-1} \sqrt{\frac{2\pi n}{(n/m)^m}} \left(\frac{1}{\sqrt{2\pi}}\right)^m \right) \\ &\stackrel{(d)}{\geq} \log \left(\frac{1}{e^{n/24}} \binom{n-1}{\frac{n}{2}-1} \frac{\sqrt{2\pi n}}{2^{n/4}} \left(\frac{1}{\sqrt{2\pi}}\right)^{n/2} \right) \\ &\stackrel{(e)}{\geq} \frac{n}{4} \log \left(\frac{4}{\pi}\right) - \frac{n}{24} \log e = \frac{n}{4} \log \frac{4}{\pi e^{1/6}} > 0.027n \end{aligned}$$

where (a) follows by the identity 14, (b) follows by Feller's inequality with a few manipulations (given in inequality (15) in [3]), (c) follows by the identity in 13, (d) follows by considering only $m = \frac{n}{2}$ term in the sum, and (e) follows by a few algebraic manipulations. ■